BISTRO Polarimetric Report: Statistical Data Passport

Automated Analysis Pipeline

14 апреля 2026 г.

Methodology and Data Integrity

This report provides a quantitative assessment of molecular cloud structures based on BISTRO survey data. A key feature of the current pipeline is the prioritized use of hardware-derived metadata for error propagation.

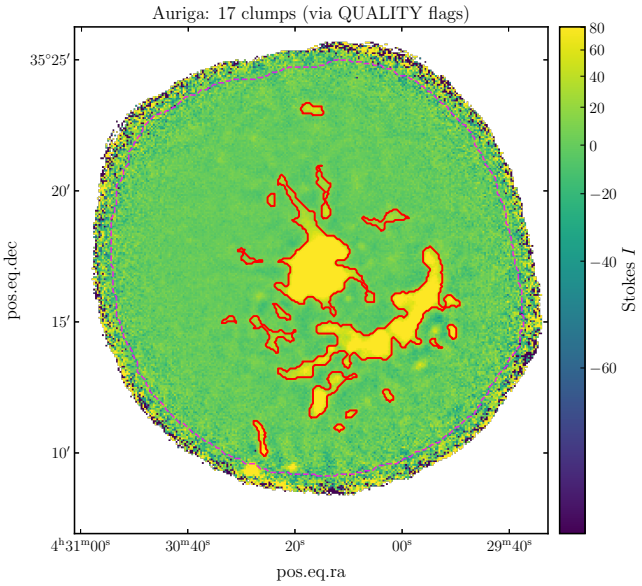
Noise Analysis Strategy: The pipeline implements a "Graceful Downgrade"logic. If the FITS headers and HDU structure (derived from the STARLINK `pol2map` reduction) contain `VARIANCE` and `QUALITY` extensions, these are utilized as the primary source of per-pixel uncertainty. This accounts for the non-uniform noise distribution inherent in the JCMT Daisy Scan pattern. In the absence of such layers, the system reverts to a heuristic estimation of the RMS within a geometric safety zone, utilizing iterative sigma-clipping to isolate the background from the emission.

Debiasing and Physics: To counteract the systematic overestimation of polarized intensity in low SNR regimes (Ricean bias), we apply the Wardle & Kronberg (1974) debiasing technique: $PI_{db} = \sqrt{PI^2 - \sigma_{PI}^2}$. Pixels failing the $SNR_{PI} > 2$ criterion are excluded from the structure function analysis to prevent noise-driven artifacts in the magnetic field morphology.

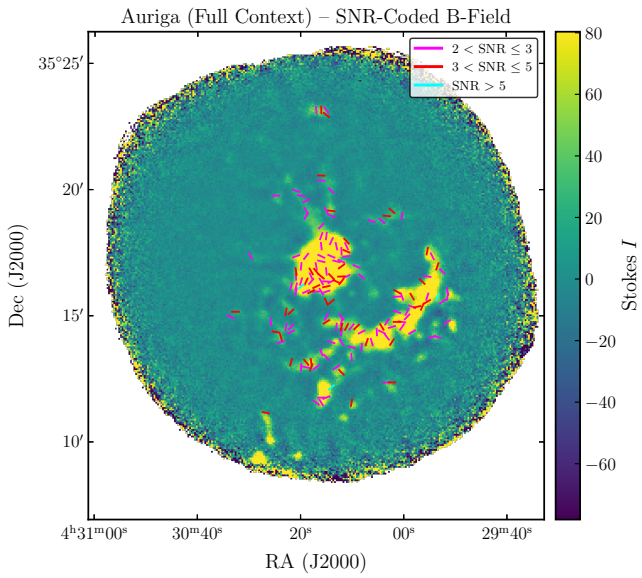
Target: Auriga

Physical Parameters and Noise Methodology

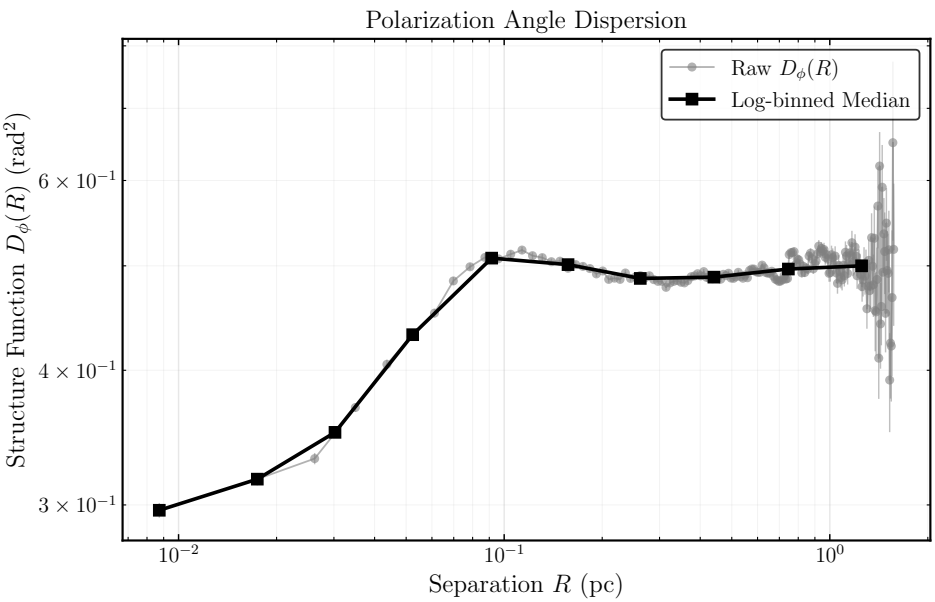
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	1502
R_range_pc	0.0087 – 1.5708
Distance (pc)	450.00000
R_range_px	[1, 180]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

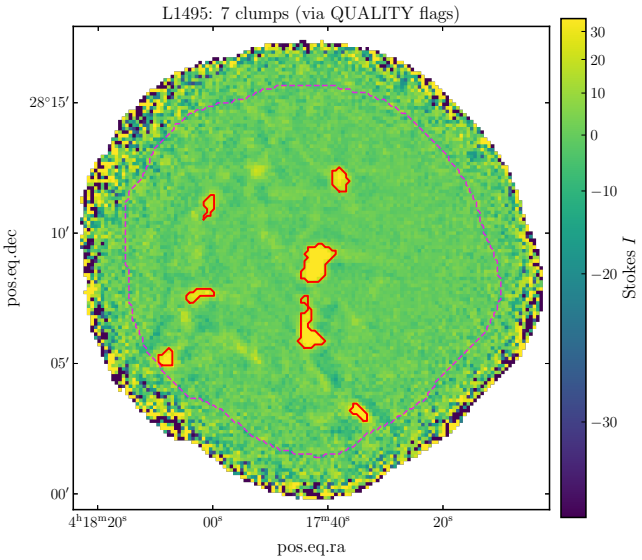


(c) Structure Function $D_\phi(R)$

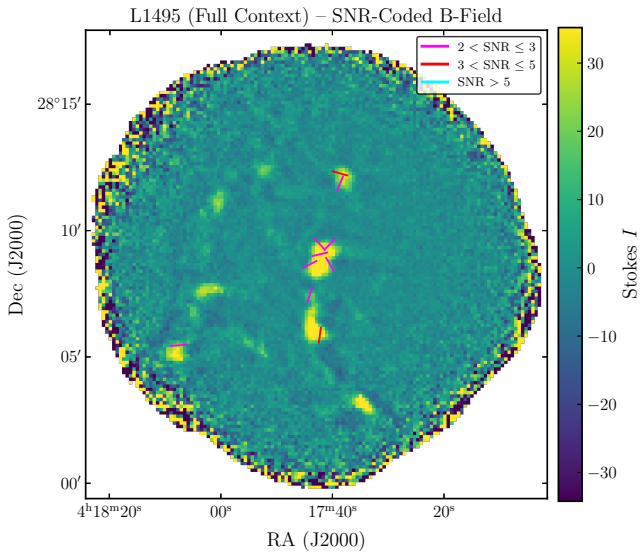
Target: L1495

Physical Parameters and Noise Methodology

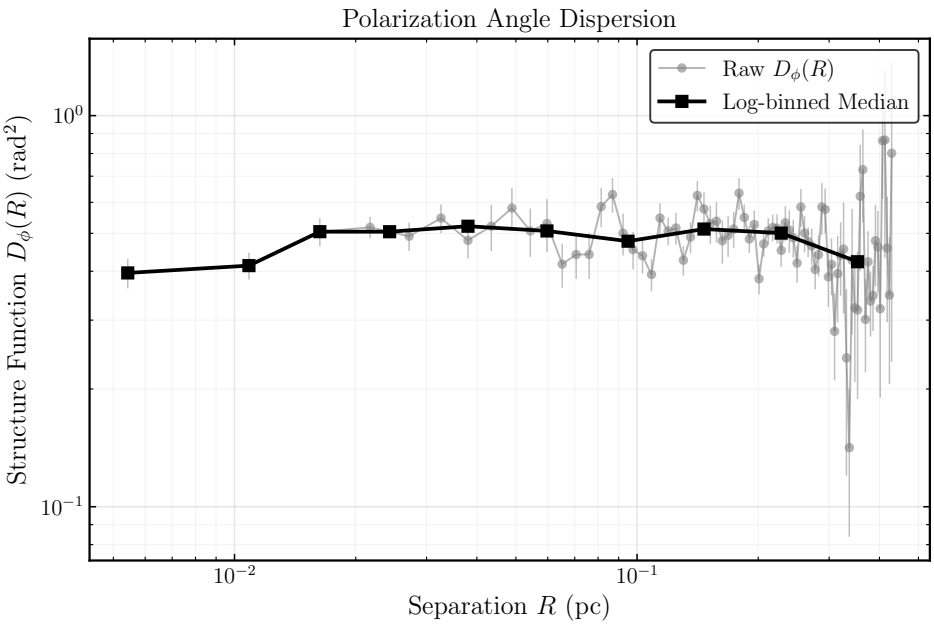
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	78
R_range_pc	0.0054 – 0.4941
Distance (pc)	140.00000
R_range_px	[1, 91]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

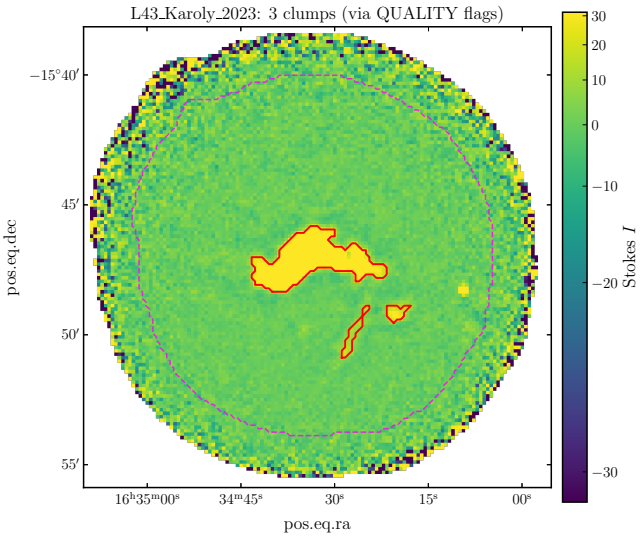


(c) Structure Function $D_\phi(R)$

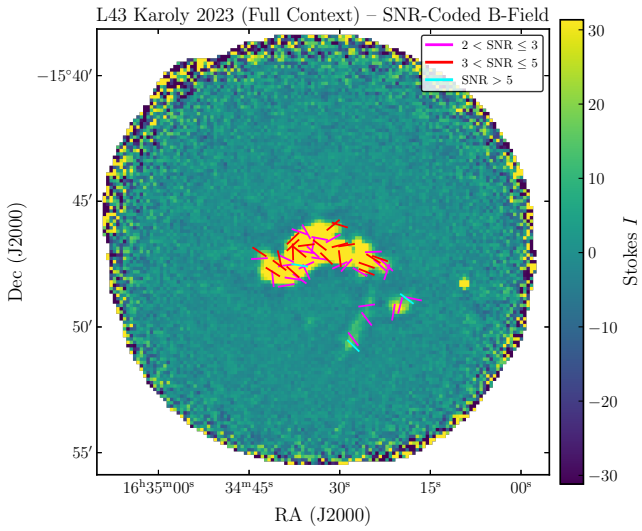
Target: L43_Karoly_2023

Physical Parameters and Noise Methodology

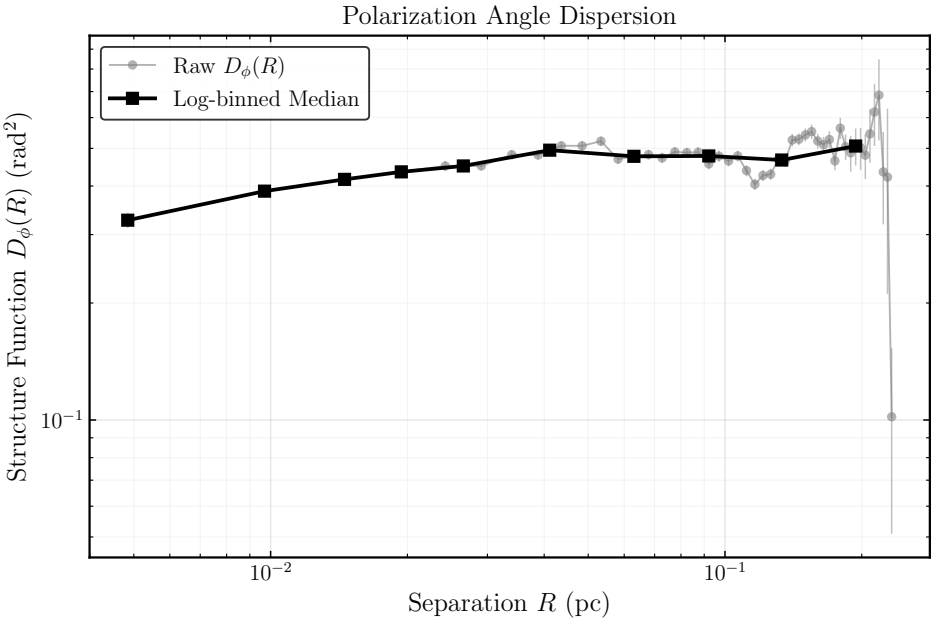
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	189
R_range_pc	0.0048 – 0.4266
Distance (pc)	125.00000
R_range_px	[1, 88]
Vector Decimation Step	2 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

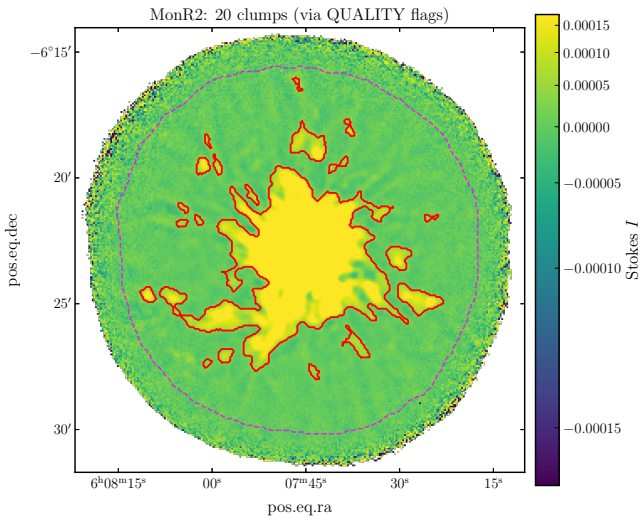


(c) Structure Function $D_\phi(R)$

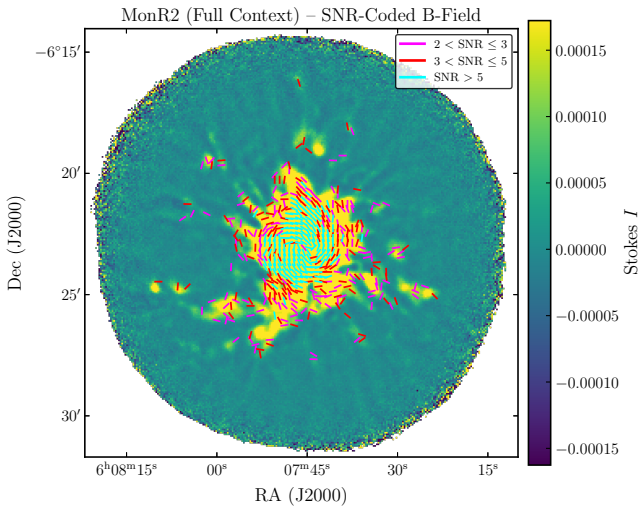
Target: MonR2

Physical Parameters and Noise Methodology

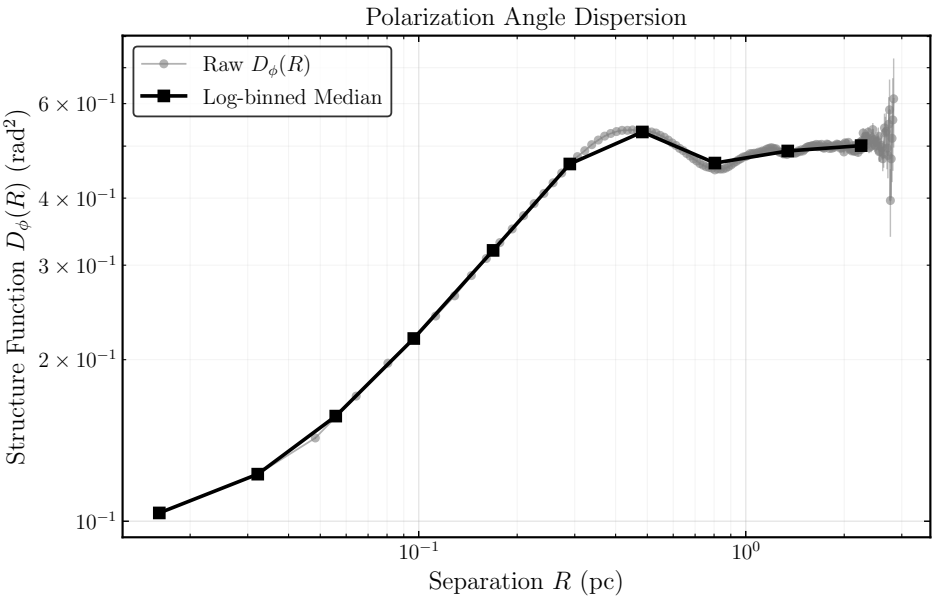
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	4402
R_range_pc	0.0161 – 2.8329
Distance (pc)	830.00000
R_range_px	[1, 176]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

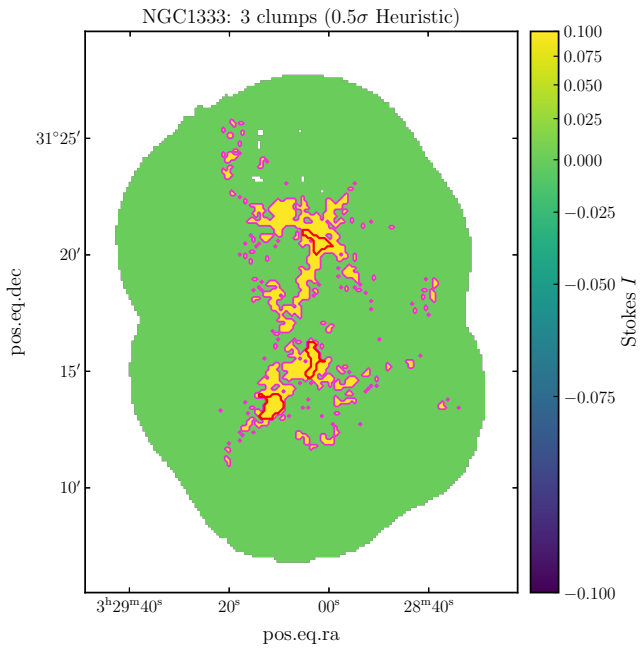


(c) Structure Function $D_\phi(R)$

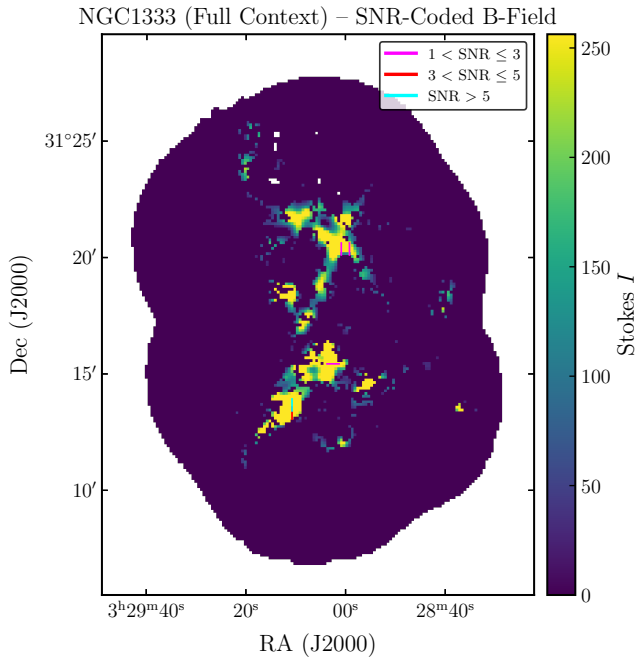
Target: NGC1333

Physical Parameters and Noise Methodology

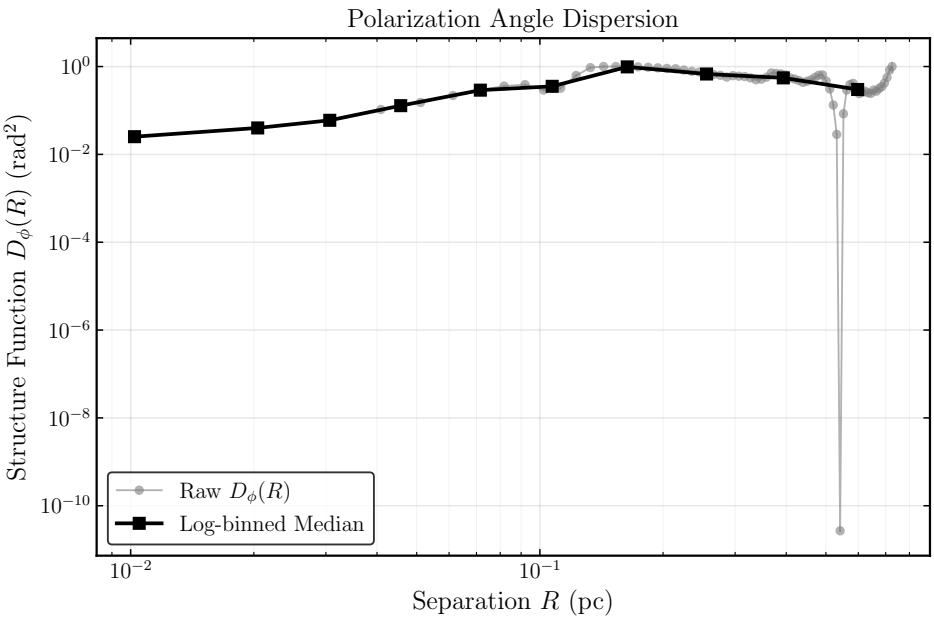
Parameter	Value
Distance	299.0 pc
Data Status	Stats JSON missing (Heuristic Fallback implied)
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

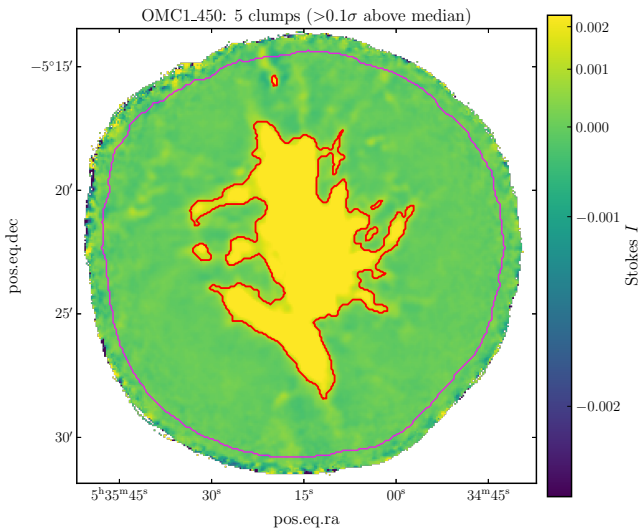


(c) Structure Function $D_\phi(R)$

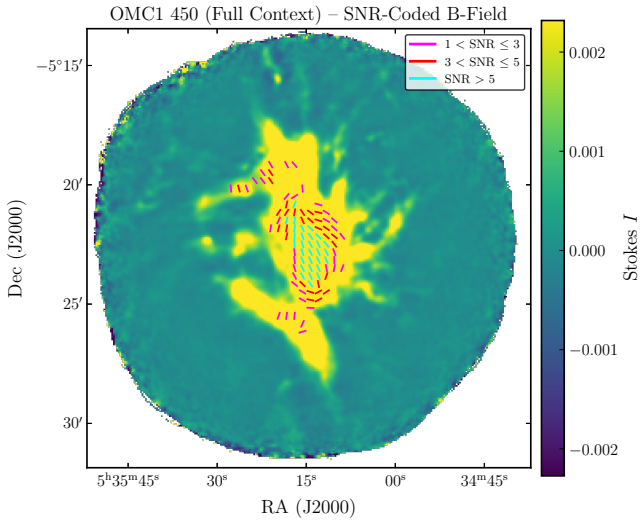
Target: OMC1_450

Physical Parameters and Noise Methodology

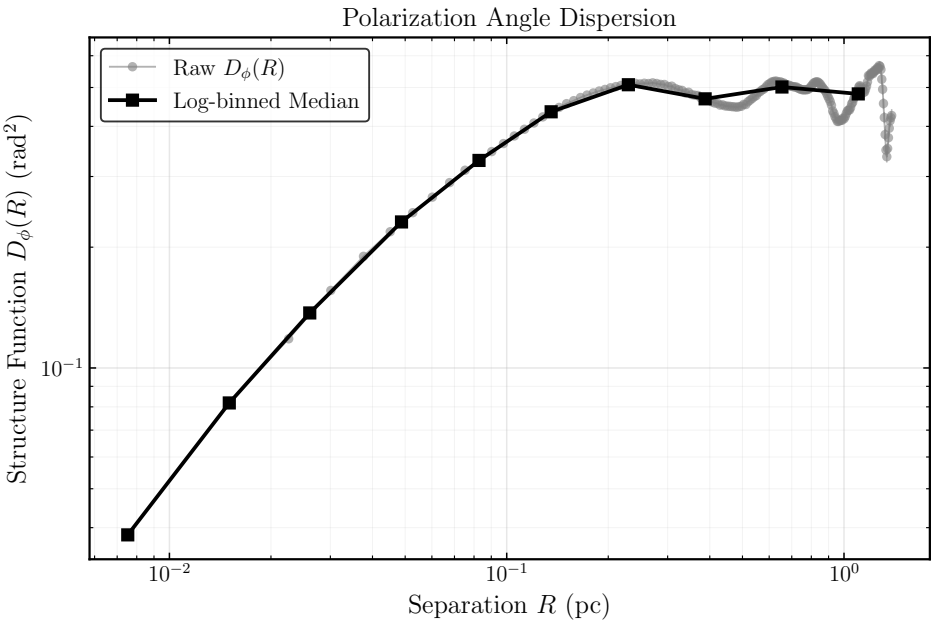
Parameter	Value
Distance	388.0 pc
Data Status	Stats JSON missing (Heuristic Fallback implied)
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

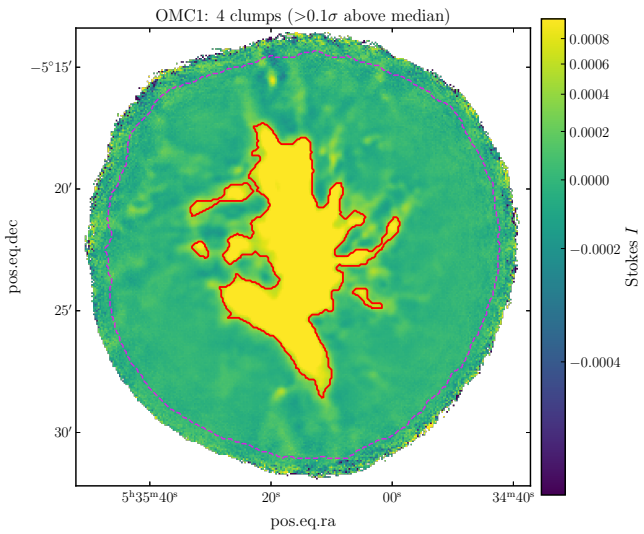


(c) Structure Function $D_\phi(R)$

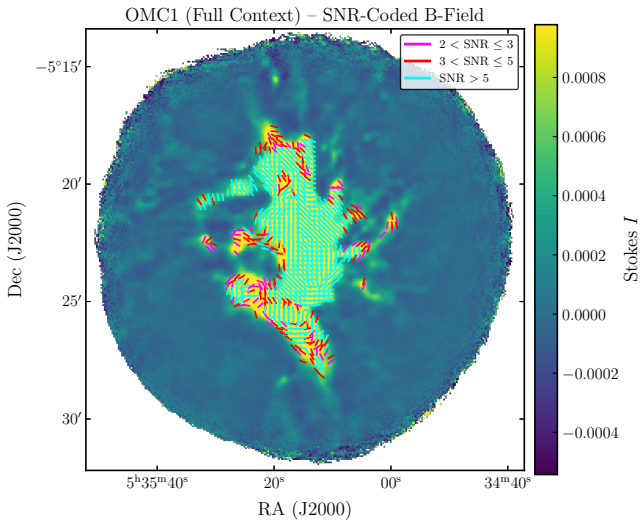
Target: OMC1

Physical Parameters and Noise Methodology

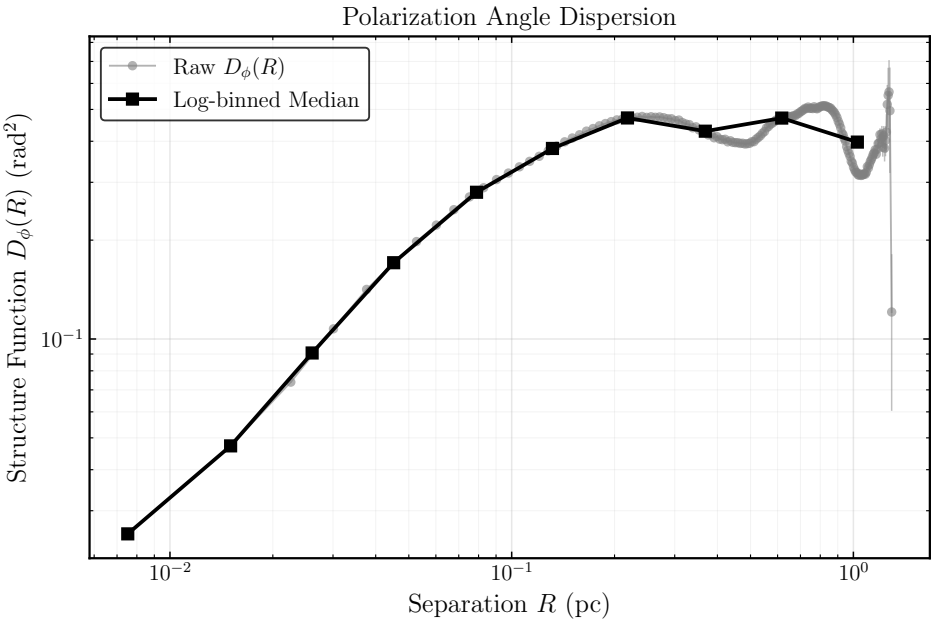
Parameter	Value
Method I	0.1σ Heuristic + Hardware Variance
Method QU	Hardware Variance
Effective Pixels	6779
R_range_pc	0.0075 – 1.3995
Distance (pc)	388.00000
R_range_px	[1, 186]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

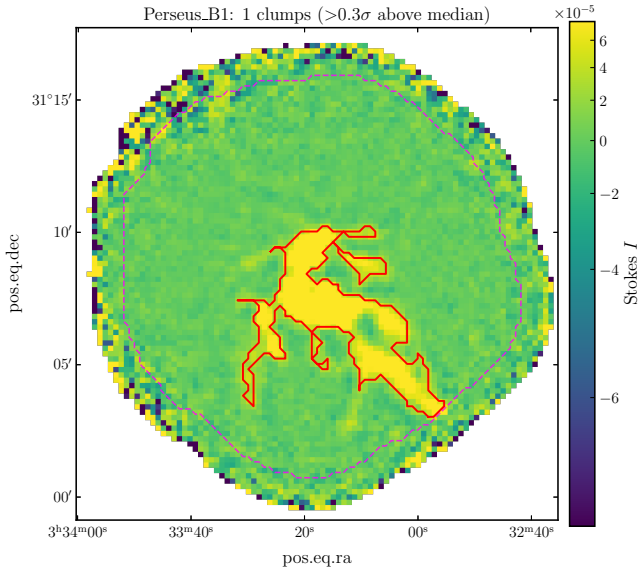


(c) Structure Function $D_\phi(R)$

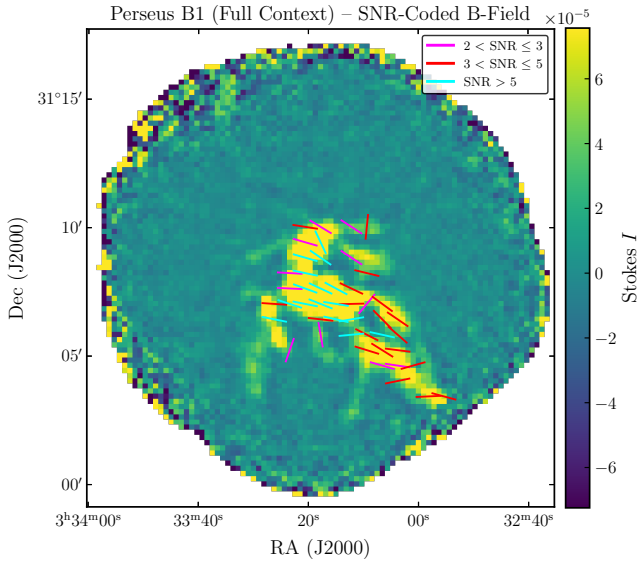
Target: Perseus_B1

Physical Parameters and Noise Methodology

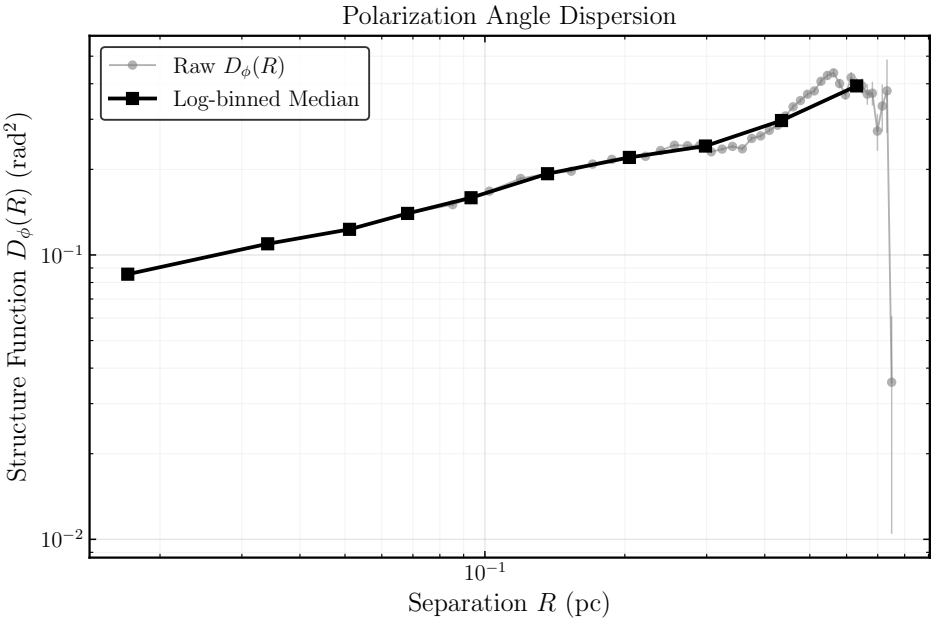
Parameter	Value
Method I	0.3σ Heuristic + Hardware Variance
Method QU	Hardware Variance
Effective Pixels	401
R_range_pc	0.0170 – 1.0228
Distance (pc)	293.00000
R_range_px	[1, 60]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

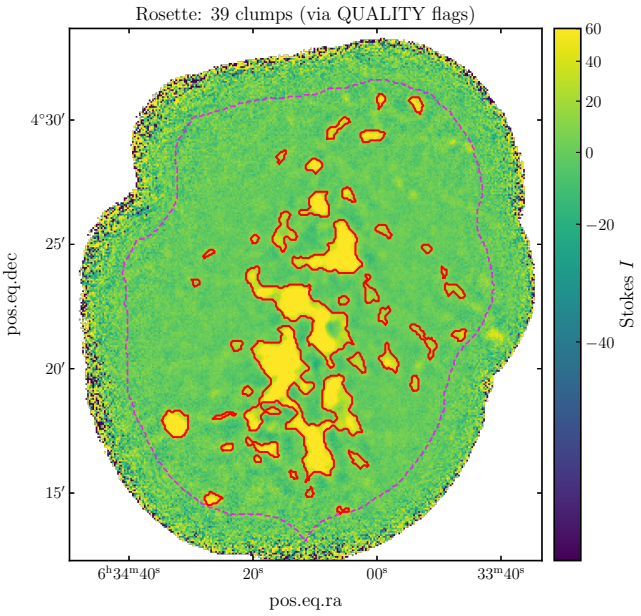


(c) Structure Function $D_\phi(R)$

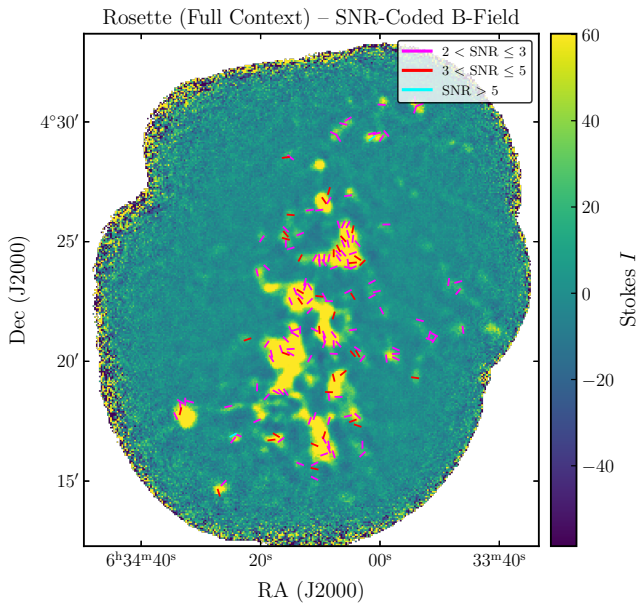
Target: Rosette

Physical Parameters and Noise Methodology

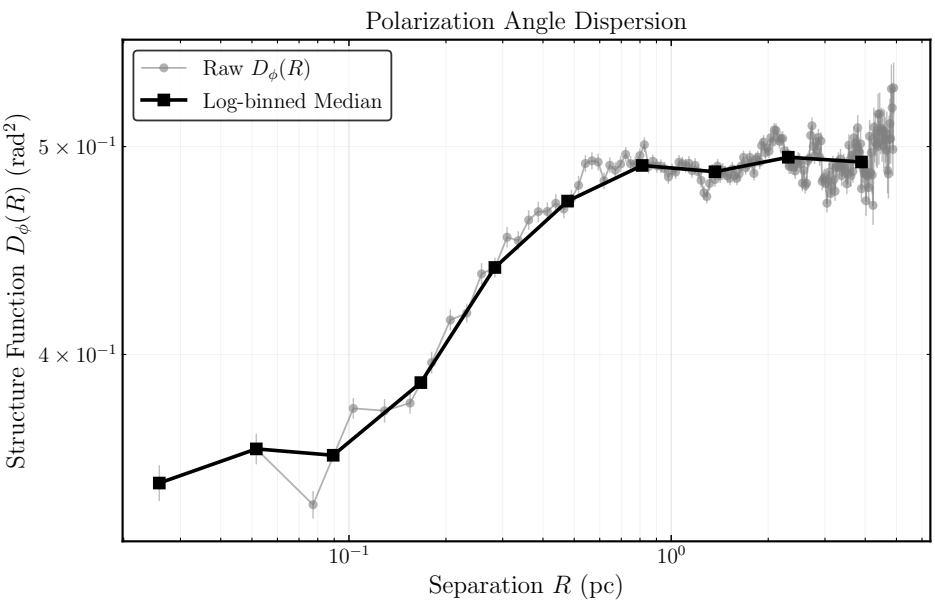
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	1452
R_range_pc	0.0258 – 4.9005
Distance (pc)	1330.00000
R_range_px	[1, 190]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

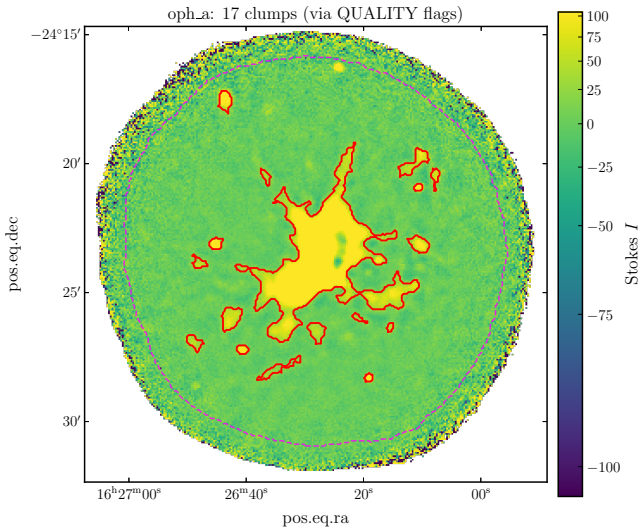


(c) Structure Function $D_\phi(R)$

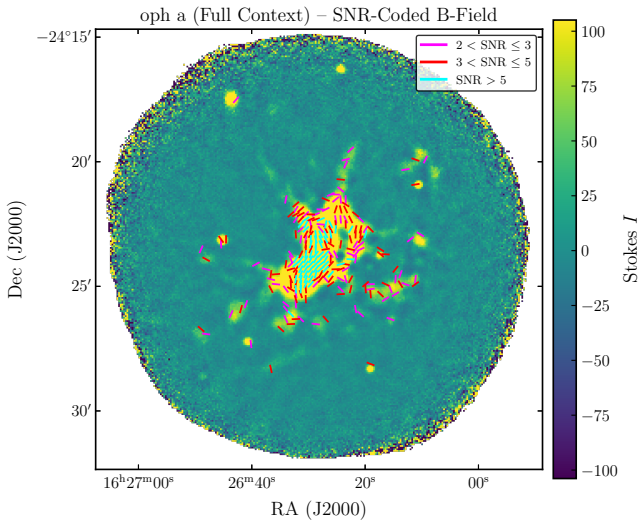
Target: oph_a

Physical Parameters and Noise Methodology

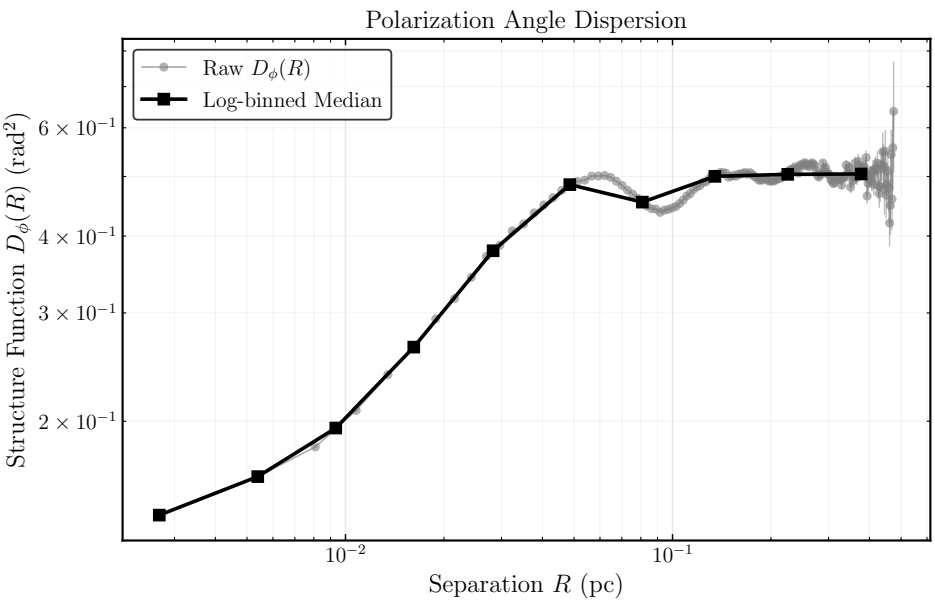
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	2288
R_range_pc	0.0027 – 0.4744
Distance (pc)	139.00000
R_range_px	[1, 176]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

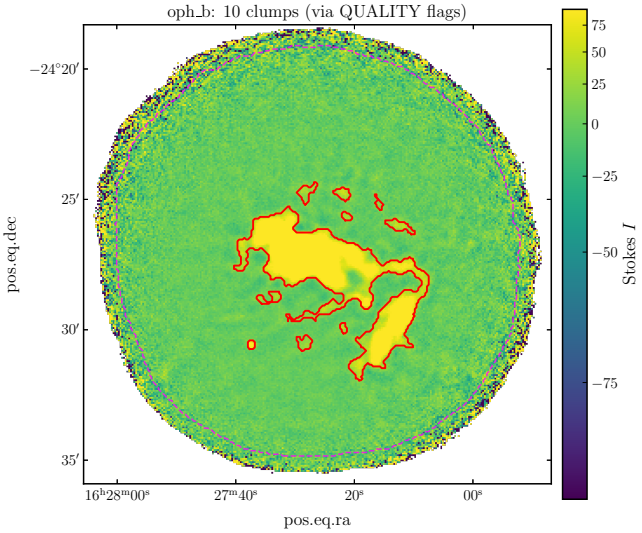


(c) Structure Function $D_\phi(R)$

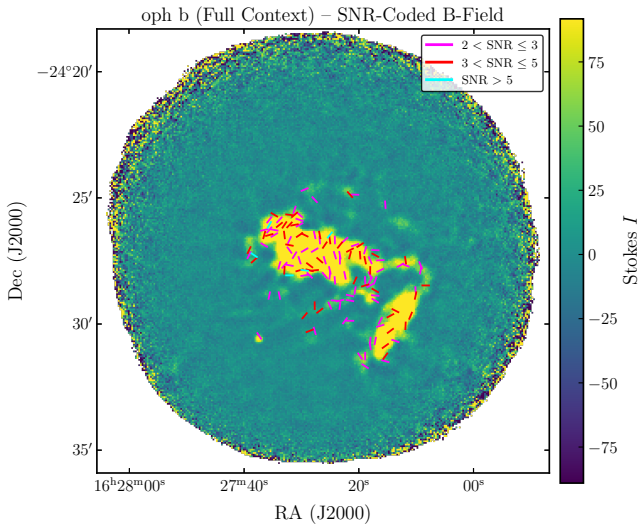
Target: oph_b

Physical Parameters and Noise Methodology

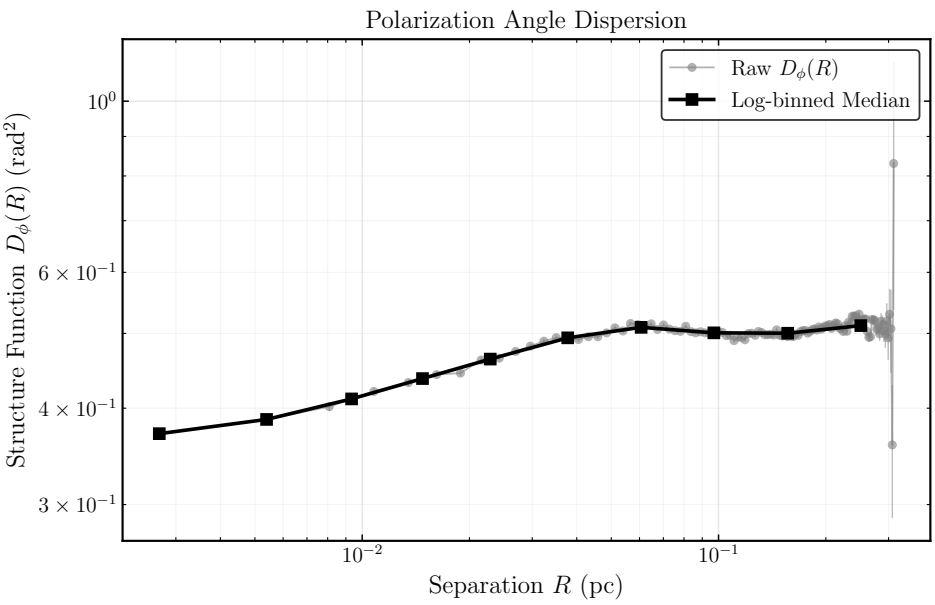
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	1370
R_range_pc	0.0027 – 0.4744
Distance (pc)	139.00000
R_range_px	[1, 176]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors

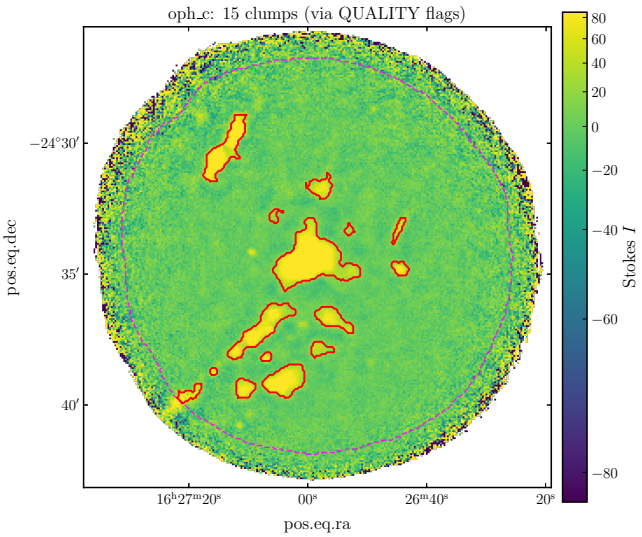


(c) Structure Function $D_\phi(R)$

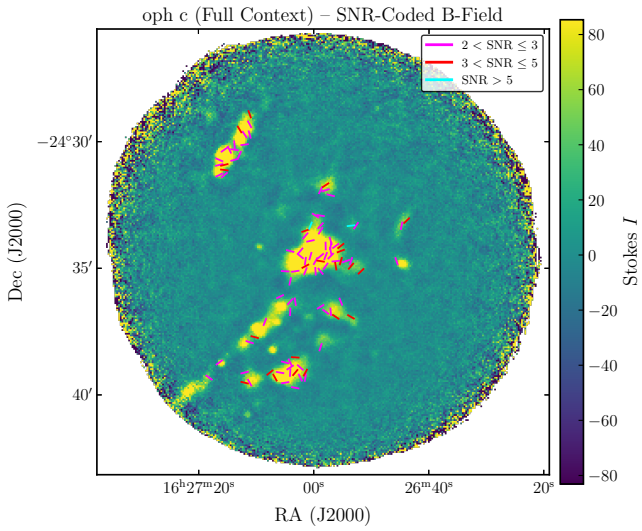
Target: oph_c

Physical Parameters and Noise Methodology

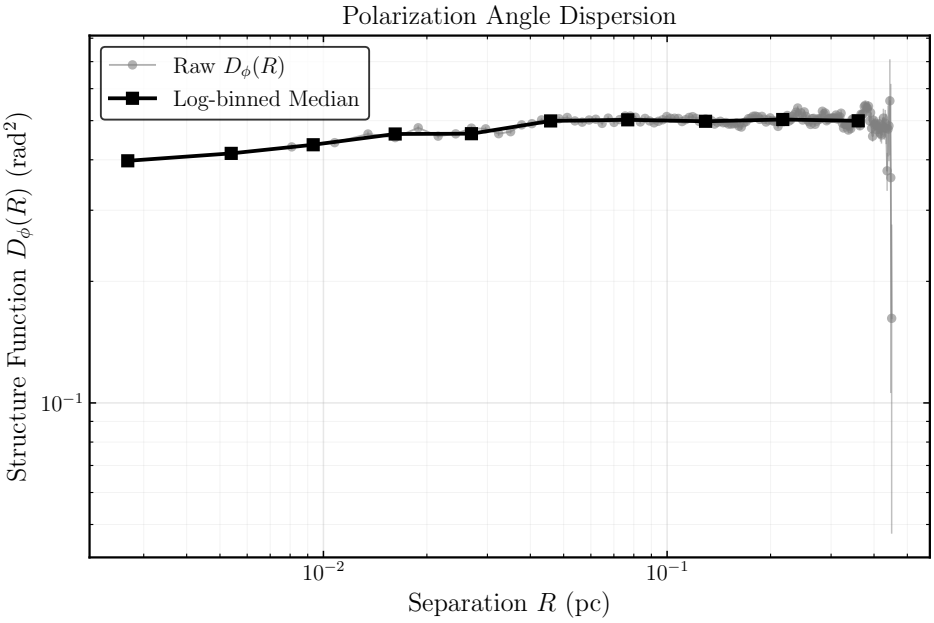
Parameter	Value
Method I	Hardware QUALITY Mask
Method QU	Hardware Variance
Effective Pixels	921
R_range_pc	0.0027 – 0.4744
Distance (pc)	139.00000
R_range_px	[1, 176]
Vector Decimation Step	3 px



(a) Segmentation & Clump Masking



(b) SNR-Coded B-Field Vectors



(c) Structure Function $D_\phi(R)$